

- 4** Parkinson's disease is caused by a loss of neurones in part of the brain called the substantia nigra.

Co-beneldopa is used to treat Parkinson's disease.

Co-beneldopa is a mixture of L-dopa and a drug that prevents the conversion of L-dopa into dopamine in the blood.

- (a) Explain why using Co-beneldopa will be more effective than using L-dopa alone to treat Parkinson's disease.

(3)

Three genes were used in this investigation:

- a control gene that is not involved in the production of L-dopa
- MYB12, a plant gene that produces a transcription factor
- CYP76, a gene found in beetroot that codes for an enzyme that converts tyrosine into L-dopa.

The table shows the concentration of L-dopa in the tomato fruit from genetically modified plants.

Genes used to modify the tomato plants	Concentration of L-dopa in the tomato fruit / mg 100 g ⁻¹
none	1.2
control gene	1.2
MYB12	1.5
CYP76	10.2
CYP76 and MYB12	14.6

(i) Explain the effects of these three genes in this investigation.

(4)

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